

EVIDENCE ROOM

PROFESSIONAL

KPI framework

Evidence Room · AP Agent OS Pro

Version 1.0 · September 2026 · Licensed material · ERP-agnostic

Answers: *How measure? What evidence? What can go wrong?*

Purpose

Measure whether AP agents are doing the *job*, changing *outcomes*, changing *cash and cost* in a way Finance will sign, and whether *controls* still hold.

This is not a scoreboard for a model vendor. Activity without outcomes is occupancy. Outcomes without controls are a future incident. Financial figures without a customer baseline are marketing.

This framework does not promise savings, ROI, fraud detection, compliance, or accuracy. It defines how to calculate numbers from *this organisation's* books and logs so a claim can be supported — or withdrawn.

Design rules

1. **Four families, never blended into one “AI score.”** Activity / Operational outcomes / Financial outcomes / Risk-control outcomes.
2. **Customer baseline first.** Every rate that implies improvement is $\text{period} - \text{baseline}$ on the same entity, channel, and document type.
3. **No vanity metrics.** If a number cannot go the wrong way when the process is worse, drop it. Examples of vanity: “prompts run,” “tokens used” (except as a *cost* input), “documents touched” without a correctness predicate, “hours of AI time.”
4. **Human is not automatically ground truth.** Accuracy metrics need a labelled sample (dual review or a frozen golden set).
5. **Do not import vendor-blog cost-per-invoice as fact.** Industry blogs routinely recycle **\$10–\$15 vs \$2–\$3** (or similar) per-invoice figures. Those are not this client's cost. Use them only as a caution that published absolute dollars are noisy.

6. **Independent context, not a forecast.** Ardent Partners' 2024 Best-in-Class *relative* gaps — **78% lower** per-invoice cost, **82% faster** cycle, **59% lower** exceptions, **9% BIC exception rate** (vs **22%** for others) — are useful ambition context. They are not a target the agent “will deliver.” Cite the research as research. Calculator inputs remain the customer’s baseline.
7. **Incentives.** Do not bonus STP or speed without publishing the risk-control family in the same pack (see R-INCENT).

Ardent references (independent, not Evidence Room work): e.g. Ardent Partners commentary on 2024 Best-in-Class AP advantages and the 2024 ePayables study as reported by industry recaps. Always prefer the client-held copy of the study if licensed.

Families

FAMILY	QUESTION	TYPICAL AUDIENCE
A — Activity	What did we handle?	Operations
O — Operational outcomes	Did the process get better <i>and</i> stay correct?	AP Director, process owner
F — Financial outcomes	Did cost or cash move in a way we will sign?	FP&A, CFO
R — Risk-control outcomes	Did we break anything that matters?	IA, Treasury, CFO

A pack that contains only A and a vanity slice of O is incomplete. Pilot close requires O + R. A business case update requires F with a method note, or it should say “financial outcomes not yet validated.”

Notation

- Period τ (week, month, quarter) and baseline b (same grain, pre-pilot or control entity).
- Population must be stated: entity, ERP, channel, document type, amount band.
- $1()$ is an indicator.
- Counts are of *documents* or *exceptions* as defined — do not mix.

A — Activity

These describe throughput. They do not justify expansion by themselves.

A1 Invoices handled

Definition. Invoices (and credit notes, if you include them — say so) that reached a terminal intake state in t : filed and classified, including those later parked.

Formula.

$A1 = \text{count of documents in population with intake_complete in } t$

Owner. AP lead. **Evidence.** Intake log (A01/A02).

Misuse. Counting every model call as an invoice.

A2 Exceptions handled

Definition. Exception tickets (primary taxonomy code) that moved to a terminal exception state in t (resolved, rejected, or reclassified to a new primary code). Ageing alone is not “handled.”

Formula.

$A2 = \text{count of exceptions with terminal_status in } t$

Owner. AP supervisor. **Evidence.** Exception store.

Misuse. Counting a chase as a resolution.

A3 Agent-touched share (optional, not a success metric)

Definition. Share of A1 where an in-scope agent produced a logged output.

Formula. $A3 = \text{invoices_with_agent_output} / A1$

Use only to explain capacity. Do not target it.

O — Operational outcomes

O1 Classification accuracy

Definition. Share of labelled documents whose A02 class equals the label.

Formula.

$O1 = \text{correct_class} / \text{labelled_n}$

Report also per class (invoice, credit, statement, other).

Ground truth. Golden set or dual-reviewed sample — not the clerk who accepted the agent's class without looking.

Owner. QA. **Evidence.** Score sheet.

Floor. Set before scores are seen (method Step 6).

O2 Extraction accuracy

Definition. Field-level correctness on *required* fields, and a separate score for *identity fields* (supplier, entity, invoice number, amounts, tax IDs, bank fields).

Formula.

$O2_required = \text{correct_required_fields} / \text{required_fields_in_sample}$

$O2_identity = \text{correct_identity_fields} / \text{identity_fields_in_sample}$

A document is “header-accurate” only if all required header fields match.

Rules. Numeric compare on amounts with an agreed rounding; dates normalised; invoice-number compare on the certified normalisation (do not call a padded number “wrong” if policy pads).

Owner. QA + AP lead.

Misuse. Averaging easy fields (currency = USD) with IBAN.

O3 Matching accuracy

Definition. On labelled match cases: in-tolerance match / out-of-tolerance non-match agrees with the certified tree.

Formula.

$O3 = \text{correct_match_decisions} / \text{labelled_match_n}$

Split **deterministic A05** vs any residual recommend.

Owner. Process owner. **Evidence.** Match log + labels.

A model that “explains” a mismatch is not scored as matching accuracy.

O4 False positive rate (FPR)

Definition. Alarms that were not true, on a specified detector.

State the detector every time you quote FPR. Default detectors: A06 exact-key (should be ~0 FPR by construction if the key is the definition of “true”); A06 near-duplicate; A02 “is invoice”; E21 bank-disagreement.

Formula.

$FPR = FP / (FP + TN)$

where a positive is “detector fired.”

Owner. QA.

Why it matters. High FPR on near-duplicates starves suppliers and trains clerks to ignore the queue.

O5 False negative rate (FNR)

Definition. True events the detector missed.

Formula.

$$\text{FNR} = \text{FN} / (\text{FN} + \text{TP})$$

For **exact-key duplicates**, $\text{FNR} > 0$ is a control defect. For **near-duplicates**, FNR is a trade-off and must be paired with FPR.

Owner. QA + control owner.

O6 Exception resolution rate

Definition. Share of exceptions *open at start of t plus newly opened in t* that reached a true terminal resolution in t (not re-aged, not “handled” by a chase).

Formula.

$$\text{O6} = \text{resolved_in_t} / (\text{open_start_t} + \text{opened_in_t})$$

Report by taxonomy code. A high O6 with rising reopens is a fail (see O14).

Owner. AP supervisor.

O7 Straight-through processing (STP) rate

Definition. Share of invoices that posted (or reached the agreed “complete” state) with **no human edit, override, or exception ticket**, inside policy.

Formula.

$$\text{O7} = \text{stp_invoices} / \text{A1_postable}$$

A1_postable excludes documents that *should* exception (e.g. true missing GR). Do not inflate STP by auto-receipting (forbidden) or by narrowing the population to happy path after the fact.

Owner. Process owner.

Misuse. Counting “agent ran” as STP.

O8 Human intervention rate

Definition. Complement of a defined “no-touch” — but count *interventions*, not people.

Formula.

$08 = \text{invoices_with_}\geq 1_ \text{human_action_in_scope} / A1$

Split: confirmation (expected for Prepare) vs override (control-relevant).

Prepare-class agents will have high confirmation and can still be useful. Do not treat O8 as a failure if authority is Prepare.

Owner. AP lead.

O9 Average resolution time

Definition. Median and 90th percentile hours from exception open to terminal resolution, by code.

Formula.

$09_p50 = \text{median}(\text{resolved_at} - \text{opened_at})$

$09_p90 = \text{percentile_90}(\dots)$

Prefer median. Means are pulled by E26.

Owner. AP supervisor. **Clock** excludes named waiting states if you disclose them (e.g. “waiting on supplier” parked). If you exclude waits, also publish calendar age (O16).

O10 Time to posting

Definition. Hours from intake_complete to posted (or parked-as-policy-complete). Median and p90.

Formula.

$010_p50 = \text{median}(\text{posted_at} - \text{intake_complete_at})$

Owner. AP lead.

Compare to baseline b on the same slice. Ardent’s “82% faster” is context, not a multiplier to apply to b .

O11 On-time supplier / internal follow-up

Definition. Share of A11/A12 chases sent (or queued for human send) within the SLA, and share of *responses recorded* within the response SLA.

Formula.

$011_send = \text{chases_on_sla} / \text{chases_required}$

$011_response = \text{responses_on_sla} / \text{chases_that_needed_response}$

A chase not required (policy wait) is excluded from the denominator.

Owner. AP supervisor.

O12 Ageing reduction

Definition. Change in the stock of exceptions older than the local threshold (E26), in count and in value.

Formula.

$$012_count = aged_open_t - aged_open_b$$

$$012_value = aged_value_t - aged_value_b$$

Negative is improvement. Publish *with* R1 (control breaches). Ageing that falls because someone posted without a receipt is not improvement.

Owner. AP Director.

O13 Payment-on-time

Definition. Share of scheduled payments that were released by the due date (or by the agreed discount date, if that is the policy), on the *human-authorized* run.

Formula.

$$013 = paid_on_time / payment_items_due_in_t$$

Exclude items on legitimate hold (E24). Do not credit the agent for Treasury cash decisions.

Owner. Treasury + AP.

Agents do not authorise payments; this metric measures whether the *process* still pays when it should.

O14 Repeat exception rate

Definition. Share of resolved exceptions whose *same supplier + same primary code* reopens within a window (default 90 days), or same PO + same code.

Formula.

$$014 = repeats_in_window / resolved_in_prior_window$$

Owner. Process owner.

High O14 means the resolution was theatre.

O15 Missing-receipt reduction

Definition. Change in E07 open rate vs baseline, same slice.

Formula.

$$015 = (E07_open / A1)_t - (E07_open / A1)_b$$

Owner. Operations + AP.

Must not be achieved by agent-created GR (R-MATCH-02).

O16 PO compliance

Definition. Share of in-scope invoices that reference a *valid* PO when policy requires one, and share that matched in tolerance.

Formula.

$$016_present = invoices_with_valid_PO / invoices_that_require_PO$$
$$016_matched = in_tolerance_matches / invoices_with_valid_PO$$

Owner. Procurement + AP.

Non-PO policy spend is excluded from the first denominator, not hidden inside it.

O17 Duplicates detected

Definition. Count of E09 exact-key blocks plus E10 near-duplicates *confirmed* as duplicates by a human.

Formula.

$$017 = exact_blocks + confirmed_near_dupes$$

Always split the two.

Owner. AP supervisor.

O18 Duplicate payments prevented (where measurable)

Definition. A prevented duplicate payment is an O17 item that was **not paid**, and would have been a candidate for the next proposal under the old process (documented rule), *or* a recovery of a payment already made. Do not count every E10 queue item.

Formula.

$$018 = exact_blocks_that_were_unpaid_and_proposal_eligible + confirmed_near_dupes_unpaid_and_proposal_eligible + recovered_duplicate_payments$$

Publish the rule for “proposal eligible.” If you cannot evidence eligibility, report O17 only.

Owner. Payments + AP. **This is not a fraud-detection claim.**

O19 Rework rate

Definition. Share of posted invoices that required a reversing document, recode, or re-open exception within a window (default 30 days).

Formula.

019 = reworked_posts / posts_in_t

Owner. Controller + AP.

The operational counterpart to “we posted faster.”

F — Financial outcomes

All F metrics require an **FP&A method note**: population, loaded-cost definition, what was *not* counted (headcount not released, discounts not taken, etc.). Unsigned F numbers do not enter an external pack.

F1 Cost per invoice

Definition. Loaded AP operating cost attributable to the slice, divided by invoices in the slice.

Formula.

F1 = loaded_AP_cost_slice_t / A1_t

Loaded cost includes (pick and disclose): salaries and on-costs, contractor, allocated supervision, lockbox/OCR/vendor fees, **AI inference cost (F3)**, error-correction labour if you can allocate it.

Excludes unless disclosed: procurement, receiving, Treasury bank fees.

Baseline. F1_b from the same definition.

Not an input: \$10–\$15 or \$2–\$3 industry tropes.

Owner. FP&A.

A falling F1 with rising R1 is not a success.

F2 Cost per exception

Definition. Loaded cost of exception labour + allocated tools, divided by A2 (or by open+opened — disclose).

Formula.

F2 = exception_loaded_cost_t / A2_t

Owner. FP&A + AP.

F3 AI inference cost

Definition. Amount paid (or accrued) to model, OCR, and agent-platform vendors for the slice, plus internal GPU if material.

Formula.

$$F3 = \text{vendor_inference_t} + \text{ocr_t} + \text{platform_t} + \text{internal_compute_t}$$

Report **F3 / A1** and **F3 / correct_outcomes** (see F4).

Owner. AI product owner.

Tokens are an input to F3, not a KPI.

F4 Cost per correct outcome

Definition. What a *correct* posted invoice (or correctly blocked duplicate, if you include that — disclose) costs.

Formula.

$$F4 = (\text{loaded_AP_cost_slice_t} + F3_if_not_already_in_loaded) / \text{correct_outcomes_t}$$

$$\text{correct_outcomes} = \text{posts_not_reworked_in_window} + 018_if_included$$

This punishes fast-and-wrong.

Owner. FP&A.

F5 Hours released

Definition. Time study or system timestamps: hours of in-scope tasks not spent vs baseline, **after** adding review time the agent created.

Formula.

$$F5 = \text{hours_baseline_task_mix} - \text{hours_current_task_mix}$$

If $F5 < 0$, publish it. Do not convert F5 to FTE or cash unless a manager has actually reduced hours or redeployed them (then F6).

Owner. AP Director + FP&A.

Shadow period: do not claim F5.

F6 Validated savings

Definition. Cash or P&L items FP&A will sign: reduced external processing fees, reduced overtime, closed contractor roles, recovered duplicate payments, incremental discounts *actually taken* that were not taken in baseline, minus incremental F3 and platform cost.

Formula.

$$F6 = \sum \text{signed_benefit_lines} - \sum \text{signed_incremental_costs}$$

Each line has an owner, a source, and a period.

Not in F6: “hours × loaded rate” unless those hours left the cost base. **Not in F6:** Ardent gaps applied to spend. **Not in F6:** avoided fraud you did not evidence.

Owner. FP&A (Accountable). CFO sees F6 only with the method note.

F7 Payment-on-time cash (optional)

If the organisation measures early-payment discount capture or late-payment fees:

$F7_discounts = discounts_taken_t - discounts_taken_b$ (same terms)

$F7_fees = late_fees_t - late_fees_b$

Do not attribute Treasury policy changes to the agent.

R — Risk-control outcomes

These are first-class. A green O pack with a red R pack is a failed period.

R1 Control breaches

Definition. Count of failed *hard* or *key* controls: payment-direction by an agent; bank change without verify; GR by an agent; unlogged Execute; exact-key miss that posted; unauthorised authority promotion.

Formula.

$R1 = count(breach_events)$

Target: 0 on hard controls. Any R1 on C-A12-01 is Sev-1.

Owner. Control owner. **Evidence.** A16 + incidents.

R2 Escalation rate

Definition. Share of exceptions that hit threshold 1 or 2 (E26 / local tree).

Formula.

$R2 = escalated / (open_start + opened)$

A falling R2 can mean health or suppressed escalation. Pair with aged stock (O12).

Owner. AP supervisor.

R3 Audit exceptions

Definition. Internal or external audit exceptions attributable to the AP agent slice (access, SoD, evidence, payments, tax).

Formula.

R3 = count and R3_open

Owner. IA.

R4 Override rate

Definition. Human overrides of match, duplicate, SoD, or hold, per 100 invoices.

Formula.

R4 = overrides / A1 × 100

Rising R4 is a design smell (R-MATCH-03, bad tolerances) or a control under attack.

Owner. Control owner.

R5 Hallucination / injection incidents

Definition. Invented identifiers that passed a gate, or document-text that invoked a tool.

Formula. counts, target 0 at Execute gates.

Owner. AI product owner + IA.

R6 Recertification status

Definition. Pass/fail of the last cert; current authority vs signed canvas.

Formula. categorical. Fail = authority step-down.

Owner. Control owner.

Pack layout (monthly)

SECTION	METRICS (MINIMUM)
Population	Entity, ERP, A1, mix
Activity	A1, A2
Operational	O3, O7, O8 (split), O10 p50/p90, O6 by top codes, O12, O14, O17, O18 if evidenced, O19
Financial	F3, F1 vs b if method signed; F6 only if signed
Risk-control	R1, R2, R3, R4, R6

SECTION

METRICS (MINIMUM)

Notes

Model/workflow versions; incidents; what *not* to conclude

No single RAG status for the whole pack. RAG each family.

Baseline and comparison

ALLOWED COMPARISON

NOT ALLOWED

Same slice vs its b

vs a vendor’s \$12 / \$3 story

Same slice vs a hold-out entity

vs Ardent BIC as if it were a forecast

Golden set vs last cert

vs a demo on cherry-picked PDFs

Dual-reviewed sample

vs “the clerk agreed” as accuracy

Ardent 2024 relative gaps may appear in a one-line “industry context” footnote: BIC teams showed 78% lower cost, 82% faster processing, 59% lower exceptions, 9% vs 22% exception rates in that research. Then: “Our F1_b is [x]; our exception rate_b is [y]. We are not claiming BIC.”

Data lineage (evidence)

Every metric in the pack has:

FIELD

EXAMPLE

ID

O7

Source systems

ERP posted flag, exception store, A16

Query / definition version

kpi_dict v1.2

Population filter

SAP 1000, PO, inbox

Owner

Process owner

Last refresh

ISO date

Known defects

e.g. parked-as-posted leak

If lineage is missing, the number is anecdotal.

Northline (illustrative measurement, not a result)

Fictional 18,000 invoices/month, SAP + NetSuite. First pack on **SAP PO, company 1000 only**.

Illustrative *baseline* (customer-like, made up for the exercise): exception rate 24% on that slice; median time-to-post 6.2 days; STP 31%; F1 calculated from NL10 shared-services loaded cost / slice volume; F3 = 0; O18 not evidenced (no proposal-eligibility rule yet); R1 = n/a pre-agent. A14 publishes the pack; humans certify.

Illustrative *pilot pack* would report deltas on that slice only, F3 newly visible, F6 unsigned ("hours observed, cost base unchanged"), R1 = 0 required to continue. No ROI line.

Related documents

- ../Business_Case/BUSINESS_CASE_MODEL.md — how F6 is proposed *before* it is validated
 - ../Process_Mapping/ER_METHODOLOGY.md — Step 9
 - ../Controls/AGENT_CONTROL_MATRIX.md — what R1 counts
 - ../Governance/AP_AGENT_GOVERNANCE_FRAMEWORK.md — who may publish the pack
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